Organisations working on full duplexity in radio communication

- Broadcom
- Columbia University
- Darpa
- GenXComm
- Intel Capital
- Kumu Networks
- Mixcomm
- Qualcomm
- Silicon Audio Inc.
- Stanford University
- Texas Instruments
- University of Colorado, Boulder
- University of Texas at Austin

wireless research. His current research interests include integrated devices, circuits and systems for a variety of RF, mm-wave and sub-mm-wave applications.

Various researches are being carried out across the world for full duplex communication. Various start-ups are arising around the technology around the world. These are being seed-funded under various government organisations, such as DARPA, and other venture capitalists.

A team of researchers from Department of Electrical and Computer Engineering at University of Texas at Austin (UT ECE), led by Prof. Andrea Alù, also worked on similar research. He built the first-ever circulator for sound using passive circuit components, which can be embedded on a CMOS chip. Alù is also chief technology officer, Silicon Audio Inc. This Texas-based company licensed the patented technology from UT ECE for its commercialisation in the market.

Another startup from Texas, GenXComm is working on enabling full duplex for next-generation communications infrastructure. Instead of using multiple frequencies (FDD) or time slots (TDD), its technology enables true full duplex or simultaneous transmission and reception of signals in the same channel, doubling the usage of the world's spectrum. This technology is used by various companies for testing full duplex under 5G standards.

Kumu Network is working with SIC technology that enables radios to do what previously seemed impossible—send and receive at the same time on the same or adjacent channel(s). The company has generated practical and affordable technology for the same, and is rapidly finding its way into a wide variety of wireless and wireline communication applications.

Since the advent of these technologies, applications in RF domain are continuously growing with the announcement of 5G in 2014. These have found applications in radar, LTE, defence, satellite, Wi-Fi and other wireless technologies.

Use of three-port IC circulators is finding applications in wireless and telecommunication technologies. Researchers are working on four-port circulators for applications in platforms that contain frequency-converting elements, including on-chip implementation for quantum information processing and nano-photonics. EFY